

Micro-Filter Air Purifier

Innovation dossier - 1. The idea

PROPOSITION

A concept for the real 'Filter cost / replacement (weak real signal)' friction, built with the DISTRIBUTE design move - one large system -> several smaller systems.

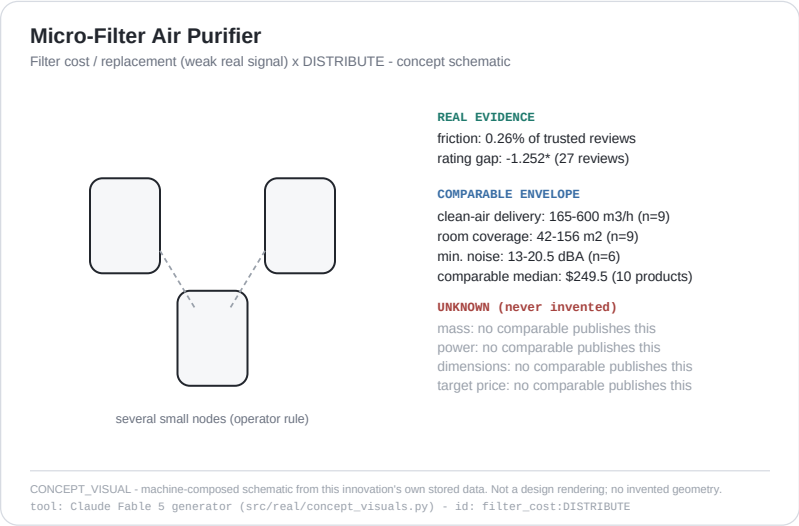


Figure: **CONCEPT_VISUAL**

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

artifact visual-filter_cost:DISTRIBUTE -
/concept-visuals/filter_cost_DISTRIBUTE.svg

ENGINEERING ENVELOPE

comparable basis: 10 verified official Versuni/Philips air-purifier families (room-scale)

metric	value from verified comparables	epistemic type
performance cadr m3h	165-600 m3/h (n=9 comparables)	OBSERVED_COMPARABLE
room coverage m2	42-156 m2 (n=9 comparables)	OBSERVED_COMPARABLE
acoustic min dba	13-20.5 dBA (n=6 comparables)	OBSERVED_COMPARABLE
target mass kg	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
target power w	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
target dimensions	No verified comparable publishes this value - left unknown, never estimated.	UNKNOWN
reference market price usd	median \$249.5 (range \$43.99-\$618.0, n=10) - market price of affected products today, not a proposed price	REFERENCE_MARKET_PRICE

State: **REJECTED**

Killed in the Magic funnel: Not enough real complaint evidence - 0.26% is below the 0.5% floor.

WHY NOW - THE OBSERVED REALITY

0.26% of trusted reviews (27 reviews across 15 real products) carry this friction, with a -1.252* average rating gap. Signal: Filter cost/frequency is a real complaint with no matching peer-reviewed literature.

TARGET USER / CONTEXT

Households whose real reviews carry this friction: 27 reviews across 15 real products (70.4% verified purchase, 2012-04-24-2023-07-21).

UNKNOWN - no persona/behavioural dataset exists in this pipeline; only the review evidence above is real.

MECHANISM - METHOD CHOICE

DISTRIBUTE - One large system -> several smaller systems.

METHOD_CHOICE - authored design vocabulary

PRODUCT ARCHETYPE - DESIGN RULE

mobility: Multi-unit - operator DISTRIBUTE turns one system into several smaller ones.

filter service dependency: High - this concept directly targets filter cost/frequency friction.

room scale, size class, power class, noise ambition, maintenance burden, connected: UNKNOWN - no real data in this pipeline for this field.

DESIGN_RULE - each populated field follows from the operator's own fixed definition, never from evidence

THE DERIVATION - THREE PARTS, EACH LABELLED

1. Reality **OBSERVED**

0.26% of trusted reviews (27 reviews across 15 real products) carry this friction, with a -1.252* average rating gap. Signal: Filter cost/frequency is a real complaint with no matching peer-reviewed literature.

2. Transformation **METHOD_CHOICE**

METHOD CHOICE, not evidence: the authored operator table assigns DISTRIBUTE - "One large system -> several smaller systems." - to this friction. Analyst design judgment, applied deterministically.

3. Product consequence **FEASIBILITY_PRECEDENT**

Feasibility precedent (from the trend corpus, not a consumer-evidence consequence): No real technical or regulatory document in this corpus addresses subscription logistics or bundled-pricing design - like customer_service, this is a real business-model change, not a technology gap, and this pipeline has no real Versuni pricing/operations capability data to assess it. Rating reflects LOW technical complexity only; commercial feasibility is genuinely unassessed here.

PARENT PATHS

No parent paths - this innovation cites no tension or assumption paths; its grounding is the friction evidence on page 1.

evidence ids: taxonomy:filter_cost

DESIGN DNA

F consumer friction (taxonomy:filter_cost) **PRESENT**

Real Amazon review-text friction theme (see Consumer Pain methodology).

O design operator **METHOD_CHOICE**

Authored operator vocabulary applied deterministically (METHOD_CHOICE, not evidence) - see OPERATORS/THEME_OPERATORS in src/real/magic_box_real.py.

S signal (filter_cost) **PRESENT**

Filter cost/frequency is a real complaint with no matching peer-reviewed literature.

T scientific tension **MISSING_UNVERIFIED**

No real research tension shares a paper with this theme's signal evidence.

R rival gap **MISSING_UNVERIFIED**

No real competitor-weakness data clears the white-space threshold for this theme.

C versuni capability **MISSING_UNVERIFIED**

No real Versuni internal capability/org-readiness dataset exists in this pipeline.

A category assumption **MISSING_UNVERIFIED**

No category assumption shares a paper with this theme's signal evidence.

E economic condition **PRESENT**

Real price-weighted exposure computed for this theme (\$7,390.66).

ECONOMICS - RELATIVE INDICATORS ONLY

price-weighted exposure: \$7390.66

SUM of real observed listed prices across affected reviews with a known price. This is a RELATIVE exposure indicator, not a revenue or market-size estimate - it has no units-sold, conversion-rate, or time-period basis. It answers 'which friction touches more expensive products' not 'how much money is at stake per year.'

comparable market median: \$249.5 (n=10 products) - not this concept's price

MEDIAN real listed price across the 10 distinct real products affected by this friction that have a known price. This is what products in this segment actually cost today - not a proposed price for a new concept, which this evidence cannot establish.

category assumptions: No category assumption shares a paper with this theme's signal evidence.

UNCERTAINTIES

- No real Versuni pricing, margin, or subscription-operations data - commercial feasibility is genuinely unassessed, not just unfavourably assessed.

CONTRADICTIONS

- none recorded for this innovation

OBSERVED VS ESTIMATED VS UNKNOWN

observed: friction stats; envelope: performance cadr m3h, room coverage m2, acoustic min dba; DNA F/S/E

method choice: operator, archetype design rule; DNA O

unknown: envelope: target mass kg, target power w, target dimensions; persona; DNA T/R/C/A

Where this idea stands: **REJECTED - GRAVEYARD**
Killed in the Magic funnel: Not enough real complaint evidence - 0.26% is below the 0.5% floor.

PROTOTYPE STATE

CONCEPT_VISUAL
CONCEPT_VISUAL means a schematic exists, nothing more: no digital model, no physical prototype, no test data.
Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

NEXT EXPERIMENT

Real internal margin/logistics data for a filter-subscription model would let this split into a proper technical-feasibility rating plus a separate, currently-missing commercial-feasibility rating.

KILL CRITERION

No authored kill criterion yet - a non-formal innovation earns one by displacing the formal recommendation (Scenario lens) or by a human decision to invest.

CRITIC DIMENSIONS

EVIDENCE	REJECT	Consumer Pain evidence-sufficiency gate failed - no real, materially-prevalent friction to build against.
ECONOMIC	CHALLENGE	Real price-weighted exposure (\$7,391) is below the corpus median (\$16,113) - a real but comparatively modest signal.
COMPETITIVE	NEEDS_EVIDENCE	No real rival-weakness data recorded for this friction theme.
HUMAN	NEEDS_EVIDENCE	No real usability/human-factors evidence exists in this pipeline for this concept.
PHYSICAL	NEEDS_EVIDENCE	No real lab/engineering feasibility data exists in this pipeline for this concept.
VERSUNI_FIT	NEEDS_EVIDENCE	No real organizational-capability assessment exists in this pipeline for this concept.
TIMING	NEEDS_EVIDENCE	No real market-timing/window evidence exists in this pipeline for this concept.
ROBUSTNESS	NEEDS_EVIDENCE	No real sensitivity/robustness study exists in this pipeline for this concept.
critic overall: REJECT		

CRITERIA RESULTS

NEEDS_EVIDENCE: 35 - PASS: 9 - CHALLENGE: 3 - N/A: 3 - KILL: 2

E3	CHALLENGE	Signal state SINGLE_SOURCE_FAMILY - real, but not yet independently corroborated.
H2	KILL	gate_passed=False
V8	CHALLENGE	Operator DISTRIBUTE may add surface area to manage - unverified.
C1	KILL	Not white space.
F3	CHALLENGE	medium

LINEAGE / PROVENANCE

generated by: src/real/innovations_real.py
magic run input sha256: 0b3f9862e0e270a295eacd709182802e981090e817f4f1072000e37aa83f150c
funnel snapshot: 73aa128628a9304d744c80cb718ecf82ed99b270b6e5d457a0198b59b6041e37
figure source: /concept-visuals/filter_cost_DISTRIBUTE.svg (generated 2026-08-31T16:09:19.526381+00:00 by Claude Fable 5 generator - src/real/concept_visuals.py)
data source: data/processed/innovations_real.json + data/processed/funnel_real.json (deterministic; no invented fields)
dossier generator: src/real/generate_innovation_dossiers.py